

Shuhan Zhang

BSc (Hons) Interactive Media, University of York
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RESEARCH INTERESTS

Game production, research through design, human-computer interaction, game narrative and systems design, tangible and embodied interaction.

EDUCATION

BSc (Hons) Interactive Media, University of York Sept 2023 - Jun 2026

Upper Second Class Honours (2:1). Relevant coursework: Mobile Interaction, AI and Its Application to Creative Practice, User Experience Design, Advanced Interaction Techniques and Technologies, Speculative Design.

RESEARCH & DESIGN PROJECTS

Revive - Lighting as a Behavioural Instrument in a 2.5D Pixel Game (*Unreal Engine 5, Lumen*)

- Designed and ran a controlled UE5 study with nine participants, using a 3x3 Latin square to test how lighting across Day, Sunset and Night conditions shapes player behaviour, under University of York ethics approval.
- Combined objective telemetry (completion time, lantern use, backtracking) with Likert ratings and found that difficulty and immersion diverge: Night was rated hardest by all participants yet most immersive by over half, showing lighting shapes spatial cognition beyond atmosphere.

Evaluating AI-Assisted Game Code Generation (*AI and Its Application to Creative Practice*)

- Built the same browser game six times with increasing reliance on Gemini 3 Pro Thinking, logging every prompt, generated file and manual repair to trace each version's behaviour to its instructions.
- Ran a six-map experiment, including one unsolvable map, and found that failures came from missing prompt constraints rather than misread spatial layout; adding one explicit constraint sentence removed the failure entirely.

Cogni-Stream / EnGram+ - Speculative BCI Critical Design (*Speculative Design*)

- Modelled a fictional subscription economy for a brain-computer-interface product, with tiered pricing, downgrade penalties and a visible status marker, turning digital-divide and technocapitalism theory into an interactive critique built in HTML/CSS and Figma.

MiniQuest - Gamifying the Break from Routine (*Mobile Interaction, Android / Kotlin*)

- Designed a native Android app that translates two evidence-based behavioural interventions into an RPG reward loop, built on an offline-first MVVM architecture with a custom accelerometer-based step detector.

THRESHOLD - Tangible AI Interaction Device (*Independent research through design, in progress*)

- Designing a tangible diagnostic device (Arduino Nano ESP32, LED ring, e-ink display) in which six life indicators decay across a conversation, asking the participant to judge whether the AI is still itself, framed against Locke's and Parfit's theories of identity.

COLLABORATIVE & APPLIED PROJECTS

Silent Night at the Haunted School (*Design Lead, Unity/C#, team of 4*)

- Wrote the narrative and led production for a first-person horror RPG, and designed a live-microphone screaming mechanic with content warnings and consent-gated audio.

Let's Supermarket (*Game & Interaction Design, Unreal Engine 5, team of 4, in development*)

- Leading design for a low-poly management sim, including a fluctuating wholesale-pricing system and a staff-morale AI.

WORK EXPERIENCE

Shanghai Sanyi Culture Communication Co. - Game Assistant, QQ Xuanwu Project Team Jul - Aug 2025

Tracked gameplay configuration data for live-service updates and logged issues found during version testing.

SKILLS

Engines & Tools: Unity, Unreal Engine 5 (Lumen, Blueprint), Figma, Maya, Arduino

Programming: C#, Kotlin, JavaScript, HTML/CSS, Python

Research Methods: research through design, speculative and critical design, controlled experimental design, qualitative and quantitative playtesting